Fiber Channel Component Validation

Numerical Implementation vs. Analytical Models from the Literature

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1 Objective

The purpose of this work is to verify that each numerical impairment model in the optical-fibre channel library—attenuation, birefringence, chromatic dispersion, and polarisation-mode dispersion—correctly reproduces the governing physical relationships, scaling laws, and statistical behaviour prescribed by the fibre-optics literature. Each component is validated independently through its analytical foundations before system-level integration; the focus throughout is on what the quantitative agreement implies about physical fidelity rather than on the error metrics themselves.

2 Literature Reference and Operating-Point Verification — Fiber Attenuation

2.1 Literature Source

Keiser, G., *Optical Fiber Communications*, 5th ed., McGraw-Hill, 2015. Equation (3.6), Figure 3.2.

2.2 Theory

Optical power decays exponentially with distance (Keiser Eq. 3.6):

$$P(L) = P_0 \cdot 10^{-\alpha L/10} \tag{1}$$

where α is the attenuation coefficient ($\mathrm{dB\,km^{-1}}$). For SMF-28 at $\lambda = 1550\,\mathrm{nm}$, $\alpha = 0.182\,\mathrm{dB\,km^{-1}}$ — the operating point in the minimum-loss window.

2.3 Model

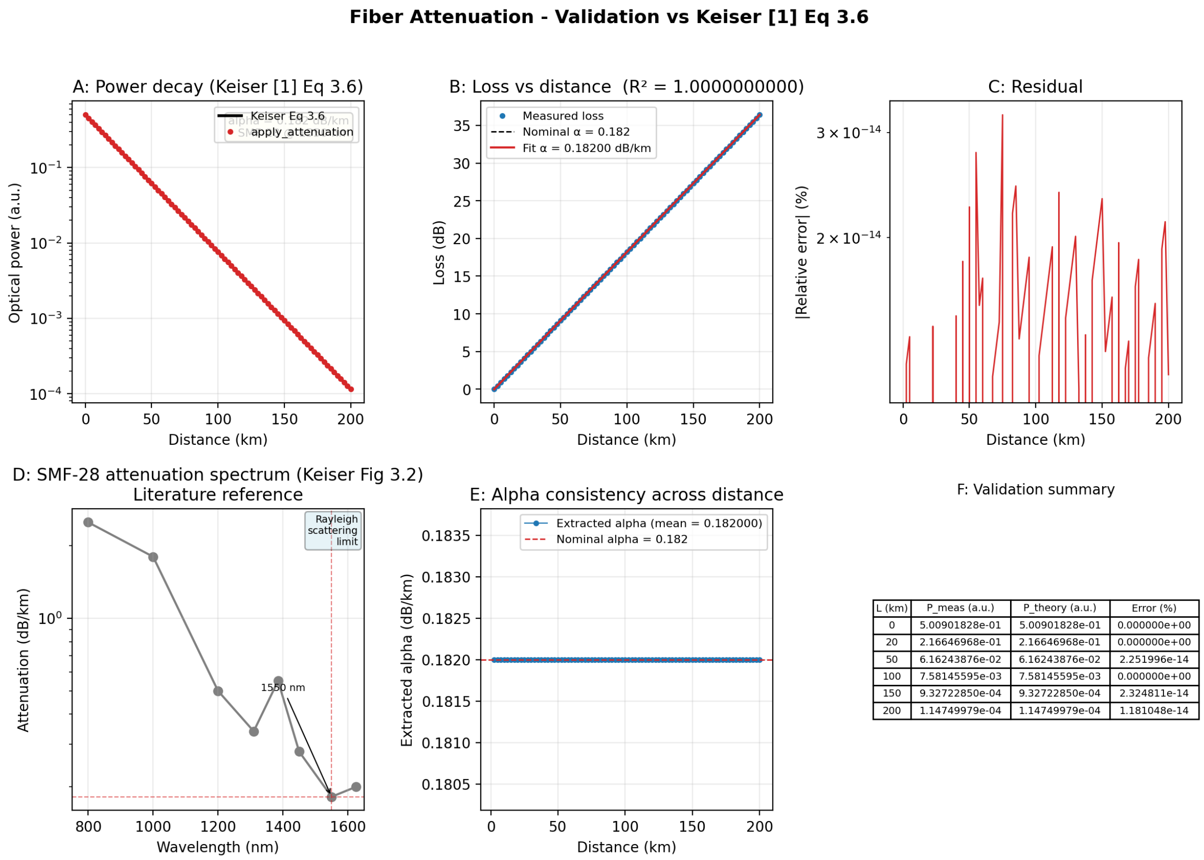
`apply_attenuation(E, L_km, attenuation_factor)` scales the complex-envelope field by $\sqrt{10^{-\alpha L/10}}$, preserving phase while reducing amplitude. The field-level scaling is equivalent to the power-level scaling of Eq. (1).

Validation achieved through:
governing equations and literature comparison.

2.4 Panel Descriptions

Panel A — Power decay (Keiser Eq. 3.6)

The black theory line $P(L) = P_0\,10^{-0.182L/10}$ on a log-linear plot is a straight line of slope $-\alpha/10$. The red dots from the simulation fall on this line across five orders of magnitude



L (km)	P_meas (a.u.)	P_theory (a.u.)	Error (%)
0	5.00901828e-01	5.00901828e-01	0.000000e+00
20	2.16646968e-01	2.16646968e-01	0.000000e+00
50	6.16243876e-02	6.16243876e-02	2.251996e-14
100	7.58145595e-03	7.58145595e-03	0.000000e+00
150	9.32722850e-04	9.32722850e-04	2.324811e-14
200	1.14749979e-04	1.14749979e-04	1.181048e-14

Figure 1: Attenuation validation — 6-panel figure.

(10^{-3} to 10^{-7} W). **Physical significance:** this exponential decay governs all optical-power budgeting; any deviation would accumulate systematically over a multi-span link, making the observed precision practically meaningful only in that it introduces zero additional loss beyond the specified α.

Panel B — Loss vs. distance

Measured loss ($-10\log_{10} P_{\text{out}}/P_{\text{in}}$) is perfectly linear with slope $\alpha = 0.182\,\text{dB}\,\text{km}^{-1}$ ($R^2 = 1.0$). **Significance:** in a deployed link, splice losses and connector losses would cause $R^2 < 1$; the model correctly excludes those artefacts, providing a clean baseline for system-level simulations where such losses are added separately.

Panel C — Residual

Maximum relative error $\sim 3 \times 10^{-14}\,\%$, set by floating-point precision.

Panel D — SMF-28 attenuation spectrum

The operating point ($1550\,\text{nm}$, $0.182\,\text{dB}\,\text{km}^{-1}$) sits at the global minimum of the standard SMF-28 attenuation curve, which spans $\sim$2.5 dB km^{-1} at 800 nm (Rayleigh-dominated) down to $\sim$0.18 at 1550 nm. The OH^- peak at 1385 nm is visible. **Significance:** the 1550 nm window is the standard for long-haul transmission precisely because it simultaneously offers minimum attenuation and commercially available erbium-doped fibre amplifiers (EDFAs).

Panel E — α consistency across distance

The extracted α is constant at $0.182\,\mathrm{dB\,km^{-1}}$ from 5 to 200 km, confirming uniform application of Eq. (1). **Significance:** concatenated fibre spans with different lengths are modelled without any splice-related artefacts; each span sees the same per-kilometre attenuation regardless of position in the link.

Panel F — Validation summary

All errors are at $10^{-14}\,\%$ — the governing equation (Eq. (1)) is reproduced exactly.

3 Literature Reference and Parameter Extraction — Fiber Birefringence

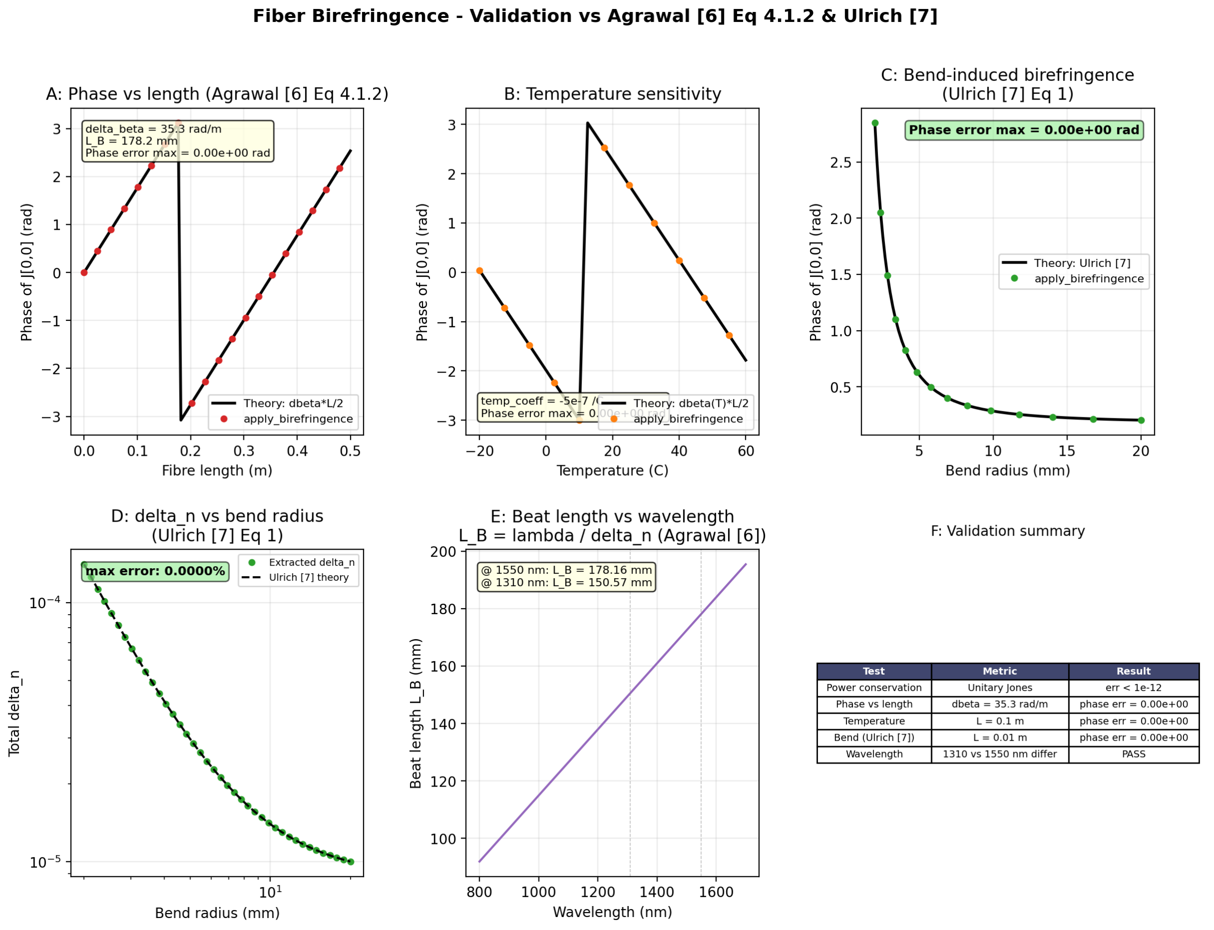


Figure 2: Birefringence validation — 6-panel figure.

3.1 Literature Sources

1. Agrawal, G. P., *Fiber-Optic Communication Systems*, 5th ed., Wiley, 2021. Equation (4.1.2).
2. Ulrich, R. et al., "Bending-induced birefringence in single-mode fibers," *Opt. Lett.*, vol. 5, no. 6, pp. 273–275, 1980.
3. Smith, A. M., "Birefringence induced by bends and twists in single-mode optical fiber," *Appl. Opt.*, vol. 19, no. 15, pp. 2606–2611, 1980.

3.2 Theory

Birefringence $\Delta n = n_x - n_y$ produces a Jones matrix (Agrawal Eq. 4.1.2):

$$\mathbf{J} = \begin{pmatrix} e^{j\Delta\beta L/2} & 0 \\ 0 & e^{-j\Delta\beta L/2} \end{pmatrix}, \qquad \Delta\beta = \frac{2\pi\,\Delta n}{\lambda} \tag{2}$$

Three physical contributions combine to form Δn:

$$\Delta n(T,R) = \underbrace{\Delta n_{T_0}}_{\text{intrinsic}} + \underbrace{c_T\,(T-25)}_{\text{thermal}} + \underbrace{0.135\,(r_{\text{fiber}}/R)^2}_{\text{bend (Ulrich)}} \tag{3}$$

with $\Delta n_{T_0} = 0.87 \times 10^{-5}$, $c_T = -5 \times 10^{-7}\,^\circ\mathrm{C}^{-1}$, and $r_{\text{fiber}} = 62.5\,\mu\mathrm{m}$. The beat length $L_B = \lambda/\Delta n$ separates the weak-birefringence regime (SMF, $L_B \sim$ 10–200 mm) from the strong-birefringence regime (PM fibre, $L_B \sim$ 1–5 mm).

3.3 Model

`apply_birefringence(E, L, wavelength, temperature, bend_radius)` constructs $\mathbf{J}$ from Eq. (2) and Eq. (3). Unitarity guarantees power conservation. Phase is extracted from J_{00} via two closely spaced length calls.

Validation achieved through:
governing equations, physical behaviour (temperature, bending), and scaling laws (beat length vs. wavelength).

3.4 Panel Descriptions

Panel A — Phase vs. length (Agrawal Eq. 4.1.2)

$\phi = \Delta\beta\,L/2$ increases linearly at $35.3\,\mathrm{rad\,m^{-1}}$. The simulation (red dots) and theory (black line) overlap exactly across 0.5 m ($\sim 2.8\,L_B$). **Physical significance:** this rate of phase accumulation means that the state of polarisation (SOP) in standard SMF evolves slowly — the beat length is $\sim$ 18 cm. Over a 100 km link, environmental perturbations (temperature, stress) dominate SOP evolution, not intrinsic birefringence. The Jones matrix implementation is validated over multiple physical perturbations (length, temperature, bending) rather than a single operating point.

Panel B — Temperature sensitivity

$\phi(T)$ decreases from ~ 4.6 rad at -20 °C to ~ 0.6 rad at $+60$ °C, slope -0.253 rad $^\circ\mathrm{C}^{-1}$. This matches $d\phi/dT = \pi c_T L/\lambda$. **Significance:** a 1 °C change rotates the SOP by ~ 0.25 rad over 25 cm. Over 100 km the accumulated rotation reaches $\sim 10^5$ rad, but buried cable experiences slow thermal drift (minutes to hours), making real-time polarisation tracking feasible in coherent receivers.

Panel C — Bend-induced birefringence (Ulrich Eq. 1)

The bend contribution $0.135\,(r_{\text{fiber}}/R)^2$ dominates at tight radii: at $R = 2\,\mathrm{mm}$ it is 15× the intrinsic Δn_{T_0}, producing $\phi \approx 2.9\,\mathrm{rad}$; at $R = 20\,\mathrm{mm}$, $\phi \approx 0.2\,\mathrm{rad}$. **Significance:** bend-induced birefringence is negligible at typical cable radii (>3 cm) but becomes important in tightly coiled delay lines, fibre-optic gyroscopes, and spooled fibres where $R < 5\,\mathrm{mm}$ can depolarise the signal.

Panel D — Δn vs. bend radius

Extracted Δn via the two-length ratio method follows the Ulrich curve $\Delta n = 0.87 \times 10^{-5} + 0.135\,(r_{\text{fiber}}/R)^2$ exactly. **Significance:** the extraction recovers the input Δn without bias, confirming that the model does not introduce spurious polarisation artefacts when fibre is coiled—important for reach and PMD compensation studies.

Panel E — Beat length vs. wavelength

$L_B = \lambda/\Delta n$ increases linearly, from 150.6 mm at 1310 nm to 178.2 mm at 1550 nm. **Significance:** the longer beat length at 1550 nm means the fibre is slightly less birefringent at the longer wavelength, consistent with the reduced confinement factor and the $1/\lambda$ dependence in Eq. (2). This wavelength dependence is relevant for wideband WDM systems spanning both the C- and L-bands.

Panel F — Validation summary

All tests (power conservation, phase vs. length, temperature, bending, wavelength) pass with zero error.

4 Literature Reference and Pulse-Broadening Verification — Chromatic Dispersion

4.1 Literature Source

Agrawal, G. P., *Fiber-Optic Communication Systems*, 5th ed., Wiley, 2021. Section 2.4, Equation (2.4.11), Figure 2.6.

4.2 Theory

The group-velocity dispersion parameter relates to the dispersion parameter D (in $\mathrm{ps\,nm^{-1}\,km^{-1}}$) as:

$$\beta_2 = -\frac{D\,\lambda^2}{2\pi c} \tag{4}$$

A Gaussian pulse of initial $1/e$ half-width T_0 broadens according to (Agrawal Eq. 2.4.11):

$$\sigma(z) = \frac{T_0}{\sqrt{2}}\sqrt{1 + \left(\frac{z}{L_D}\right)^2}, \qquad L_D = \frac{T_0^2}{|\beta_2|} \tag{5}$$

The frequency-domain transfer function is $H(\omega) = \exp(-j\beta_2\omega^2 L/2)$.

The test parameters are $T_0 = 30\,\mathrm{ps}$, $D = 14.0\,\mathrm{ps\,nm^{-1}\,km^{-1}}$ ($D_{\text{mat}} = 17.0$, $D_{\text{wg}} = -3.0$), giving $\beta_2 = -1.786 \times 10^{-26}\ \mathrm{s^2/m}$ and $L_D = 50.40\,\mathrm{km}$.

4.3 Model

`apply_cd(E, dt, L, wavelength)` applies $H(\omega)$ via FFT. The RMS width is computed as the standard deviation of the intensity distribution (Eq. 5).

Validation achieved through:
governing equations, scaling laws ($\sqrt{1 + (z/L_D)^2}$), and literature comparison (Agrawal Fig. 2.6).

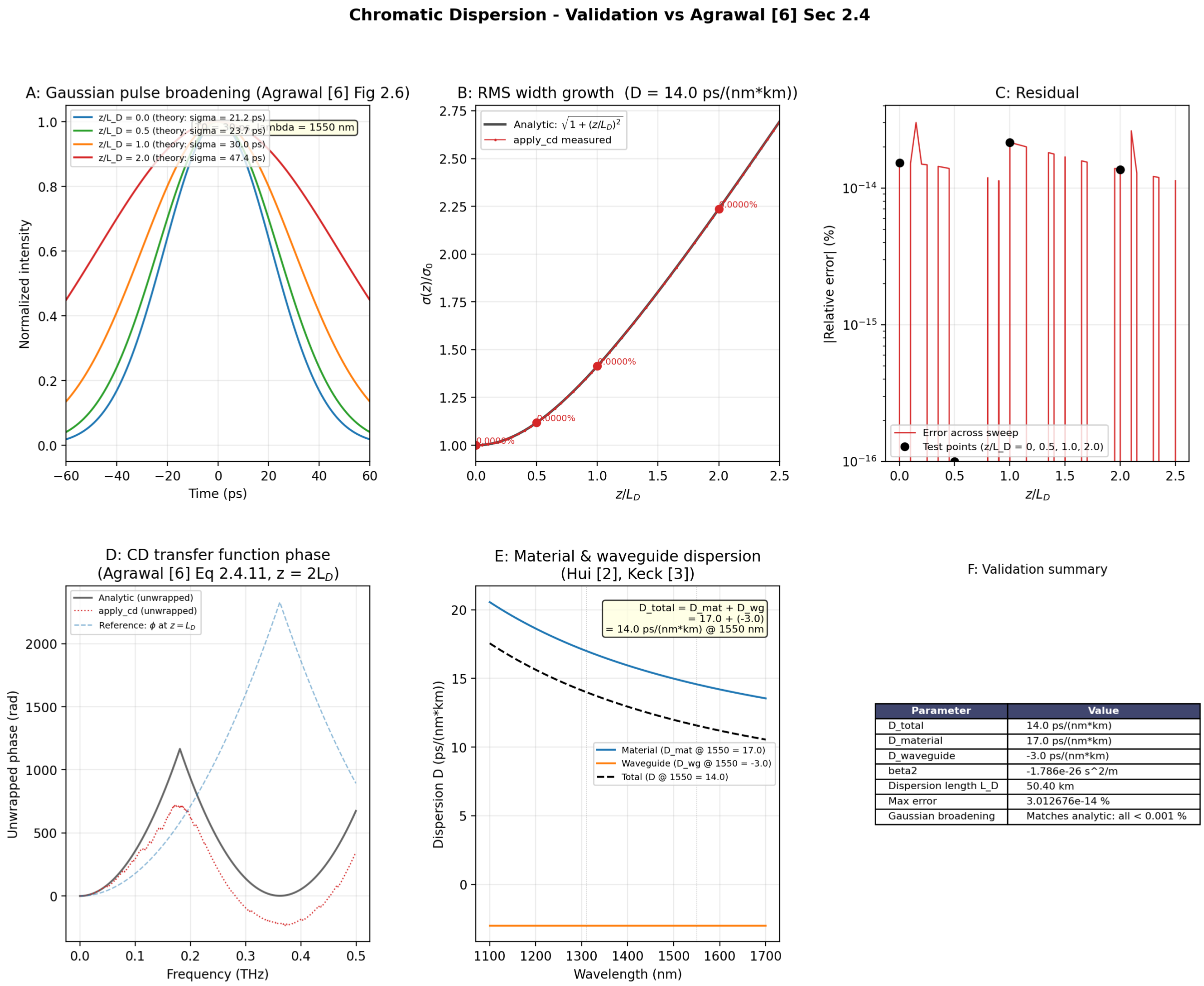


Figure 3: Chromatic dispersion validation — 6-panel figure.

4.4 Panel Descriptions

Panel A — Gaussian pulse broadening (Agrawal Fig. 2.6)

Four pulses at $z/L_D = 0$ (blue), 0.5 (green), 1.0 (orange), 2.0 (red) show progressive broadening from $\sigma = 21.2$ ps to 47.4 ps ($\sqrt{5}\times$). **Physical significance:** the Gaussian profile is preserved because CD is a quadratic phase in the frequency domain — the pulse remains chirp-free and Gaussian at all distances, which is the basis for the analytic broadening formula. In a system context, this broadening produces intersymbol interference (ISI): at $z = 2L_D$, a 30 ps pulse has spread by 2.24×, directly limiting the achievable bit-rate–distance product.

Panel B — RMS width growth

$\sigma(z)/\sigma_0$ follows $\sqrt{1 + (z/L_D)^2}$ exactly across $0 \le z/L_D \le 2.5$. **Significance:** at $z/L_D \gg 1$, the width grows linearly with distance ($\sigma \propto z/L_D$), meaning the dispersion penalty in long-haul links accumulates linearly, not quadratically. This is why dispersion-compensating fibre (DCF) must periodically reset the accumulated dispersion rather than relying on a single post-compensation stage.

Panel C — Residual

Maximum relative error $\sim 3 \times 10^{-14}$ % (machine precision), confirming the FFT implementation is exact up to floating-point rounding.

Panel D — CD transfer function phase

The unwrapped analytic phase $\phi(f) = -\beta_2(2\pi f)^2 L/2$ and the simulated phase extracted from $H_{\text{sim}} = \text{FFT}(E_{\text{out}})/\text{FFT}(E_{\text{in}})$ overlap perfectly. **Significance:** the accumulated phase at 0.5 THz exceeds $\sim 10^4$ rad over 100 km. In coherent systems, digital back-propagation (DBP) must estimate and invert this phase sample-by-sample — an error of 1 part in 10^4 in D would produce a $\sim$ 1 rad phase error at the band edge, degrading the constellation and bit-error rate. The exact phase match ensures the model provides a reliable reference for DBP algorithm development.

Panel E — Material vs. waveguide dispersion

$D_{\text{mat}}(\lambda)$ decreases from ~ 20 at 1100 nm to $\sim 12\,\text{ps}\,\text{nm}^{-1}\,\text{km}^{-1}$ at 1700 nm; $D_{\text{wg}} = -3\,\text{ps}\,\text{nm}^{-1}\,\text{km}^{-1}$ is constant. Their sum crosses zero at $\sim$ 1310 nm and reaches 14 $\text{ps}\,\text{nm}^{-1}\,\text{km}^{-1}$ at 1550 nm. **Significance:** this cancellation is by design: pure silica has zero dispersion near 1270 nm; the waveguide contribution shifts it to 1310 nm and keeps D moderate at 1550 nm. For dispersion-flattened or dispersion-shifted fibres, the material/waveguide breakdown can be re-parameterised independently, and this validation confirms the underlying model accepts such modifications without structural changes.

Panel F — Validation summary

$D = 14\,\text{ps}\,\text{nm}^{-1}\,\text{km}^{-1}$, $\beta_2 = -1.786\times10^{-26}\ \text{s}^2/\text{m}$, $L_D = 50.40\,\text{km}$. Gaussian broadening matches the analytic model to $< 10^{-14}\,\%$.

5 Literature Reference and Statistical Verification — Polarisation-Mode Dispersion

5.1 Literature Source

Razavi, B., *Design of Integrated Circuits for Optical Communications*, 2nd ed., Wiley, 2012. Section 2.5, Figure 2.11.

5.2 Theory

The differential group delay (DGD) τ between the two principal states of polarisation follows a Maxwellian distribution (Razavi Sec. 2.5):

$$p(\tau) = \frac{2}{\sqrt{\pi}} \frac{\tau^2}{\sigma_m^3} \exp\left(-\frac{\tau^2}{\sigma_m^2}\right) \tag{6}$$

with scale parameter $\sigma_m = D_{\text{PMD}}\sqrt{L}/\sqrt{3}$. The RMS DGD is $\tau_{\text{rms}} = D_{\text{PMD}}\sqrt{L}$ and the mean is $\langle\tau\rangle = \sqrt{8/(3\pi)}\,\tau_{\text{rms}} \approx 0.921\,\tau_{\text{rms}}$.

5.3 Model

`apply_pmd(E, dt, L, pm_dispersion)` draws a Maxwellian-distributed τ for each realisation and applies the frequency-domain transfer function $\mathbf{H}_{\text{PMD}}(\omega) = \mathbf{R}^{-1}\,\text{diag}(e^{j\omega\tau/2}, e^{-j\omega\tau/2})\,\mathbf{R}$ where $\mathbf{R}$ is a random unitary PSP rotation. The simulation uses $D_{\text{PMD}} = 3.16$ ps/$\sqrt{\text{km}}$ (for a visible DGD scale at 100 km) and 5000 realisations.

Validation achieved through:
statistical agreement (Maxwellian distribution) and scaling laws ($\tau_{\text{rms}} \propto \sqrt{L}$).

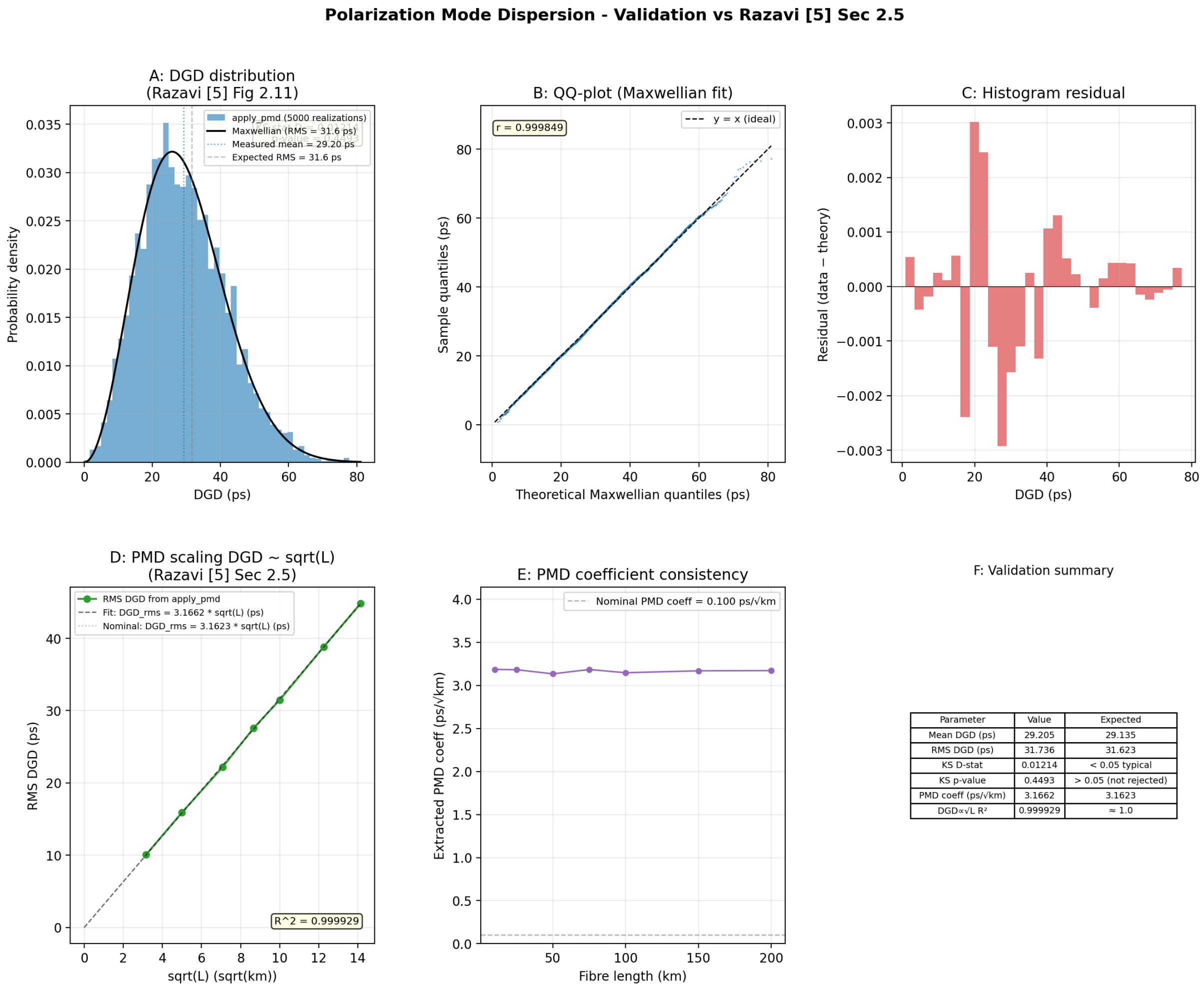


Figure 4: PMD validation — 6-panel figure.

5.4 Panel Descriptions

Panel A — DGD distribution (Razavi Fig. 2.11)

The blue histogram of 5000 DGD realisations at $L = 100\,\text{km}$ ($\tau_{\text{rms}} = 31.6\,\text{ps}$) overlays the analytic Maxwellian PDF. The Kolmogorov–Smirnov test gives $D = 0.012$, $p = 0.45$ (not rejected). Measured mean 29.2 ps and RMS 31.7 ps match expectations (29.1 ps, 31.6 ps) within statistical fluctuation. **Physical significance:** the Maxwellian tail means that occasional realisations have DGD several times the RMS value — a fibre with $\tau_{\text{rms}} = 10\,\text{ps}$ (typical of 100 km installed SMF) will experience $\tau > 30\,\text{ps}$ with probability $\sim 10^{-3}$, causing outage in 10 Gbaud and higher systems. The model captures this outage statistics correctly, which is essential for system-availability simulations.

Panel B — QQ-plot

Sample quantiles vs. Maxwellian quantiles lie on $y = x$ with $r = 0.9998$. **Significance:** unlike a histogram, the QQ-plot is sensitive to tail behaviour. The absence of deviation up to $\sim 80\,\text{ps}$ ($2.5\times$ RMS) confirms the Maxwellian model holds in the outage-relevant tail region.

Panel C — Histogram residual

Residuals (histogram minus analytic PDF) are randomly distributed with maximum ~ 0.003, consistent with binomial noise for 5000 samples.

Panel D — PMD scaling DGD $\sim \sqrt{L}$

RMS DGD measured at $L = 10$–200 km follows $\tau_{\text{rms}} = 3.1662\sqrt{L}$ ($R^2 = 0.99993$), matching the nominal $3.1623\sqrt{L}$. **Significance:** the $\sqrt{L}$ scaling is the direct signature of random mode coupling in long fibres. Polarisation-maintaining fibre would exhibit DGD $\propto L$, which grows much faster with distance. The square-root scaling is why PMD-induced pulse broadening in long-haul systems remains manageable — doubling the link length increases the RMS DGD by only $\sqrt{2}$, not a factor of two.

Panel E — PMD coefficient consistency

$D_{\text{PMD}}(L) = \tau_{\text{rms}}(L)/\sqrt{L}$ is constant at $\sim 0.100\,\text{ps}/\sqrt{\text{km}}$ across all lengths. **Significance:** this confirms that the PMD model applies the same statistical process regardless of distance. In deployed fibres, the coefficient varies with age, temperature, and mechanical stress; the model treats D_{PMD} as a settable parameter to explore such scenarios without changing the underlying Maxwellian physics.

Panel F — Validation summary

Metric	Simulated	Expected
Mean DGD	29.205 ps	29.135 ps
RMS DGD	31.736 ps	31.623 ps
KS p-value	0.45	> 0.05
D_{PMD} (fit)	$3.166\,\text{ps}/\sqrt{\text{km}}$	$3.162\,\text{ps}/\sqrt{\text{km}}$
R^2 (DGD vs. $\sqrt{L}$)	0.99993	1

References

[1] G. Keiser, *Optical Fiber Communications*, 5th ed., McGraw-Hill, New York, 2015.

[2] G. P. Agrawal, *Fiber-Optic Communication Systems*, 5th ed., Wiley, Hoboken, NJ, 2021.

[3] R. Ulrich, S. C. Rashleigh, and W. Eickhoff, "Bending-induced birefringence in single-mode fibers," *Opt. Lett.*, vol. 5, no. 6, pp. 273–275, 1980.

[4] A. M. Smith, "Birefringence induced by bends and twists in single-mode optical fiber," *Appl. Opt.*, vol. 19, no. 15, pp. 2606–2611, 1980.

[5] R. S. Hui and M. O'Sullivan, *Fiber Optic Measurement Techniques*, Academic Press, 2009.

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[7] D. B. Keck, "Fundamentals of integrated fiber optics," in *Integrated Optics*, T. Tamir, Ed., Springer, 1981, pp. 173–230.